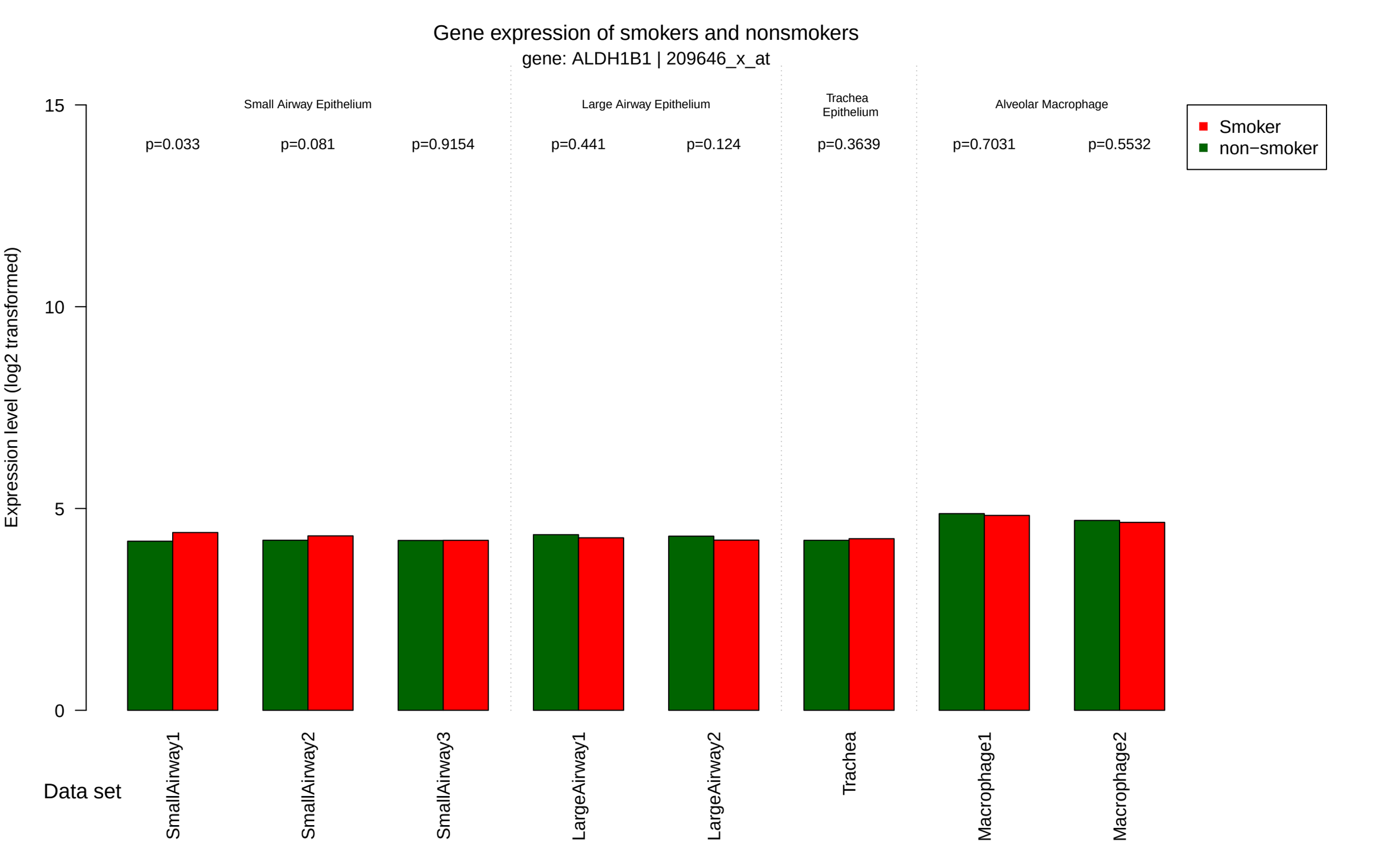
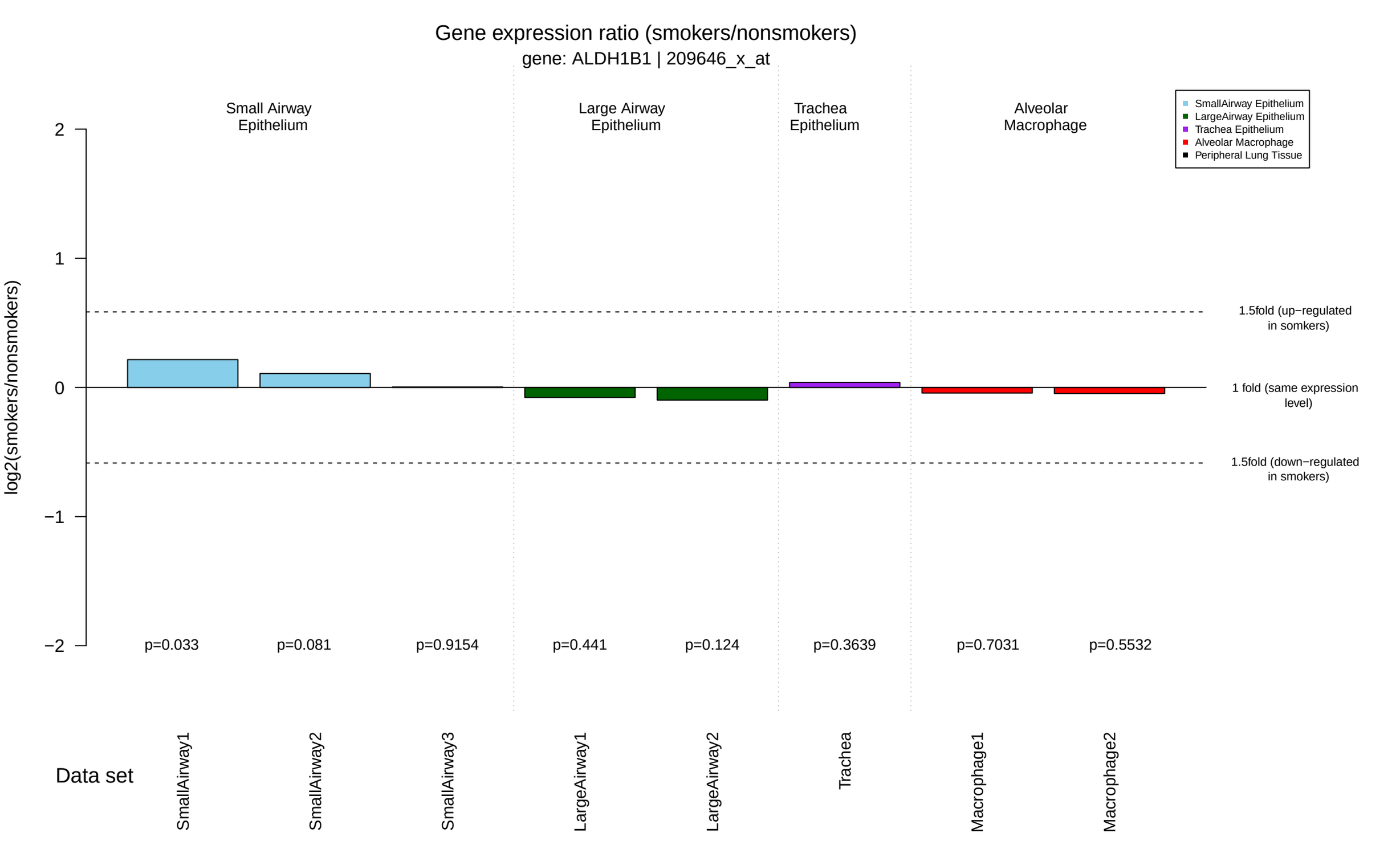
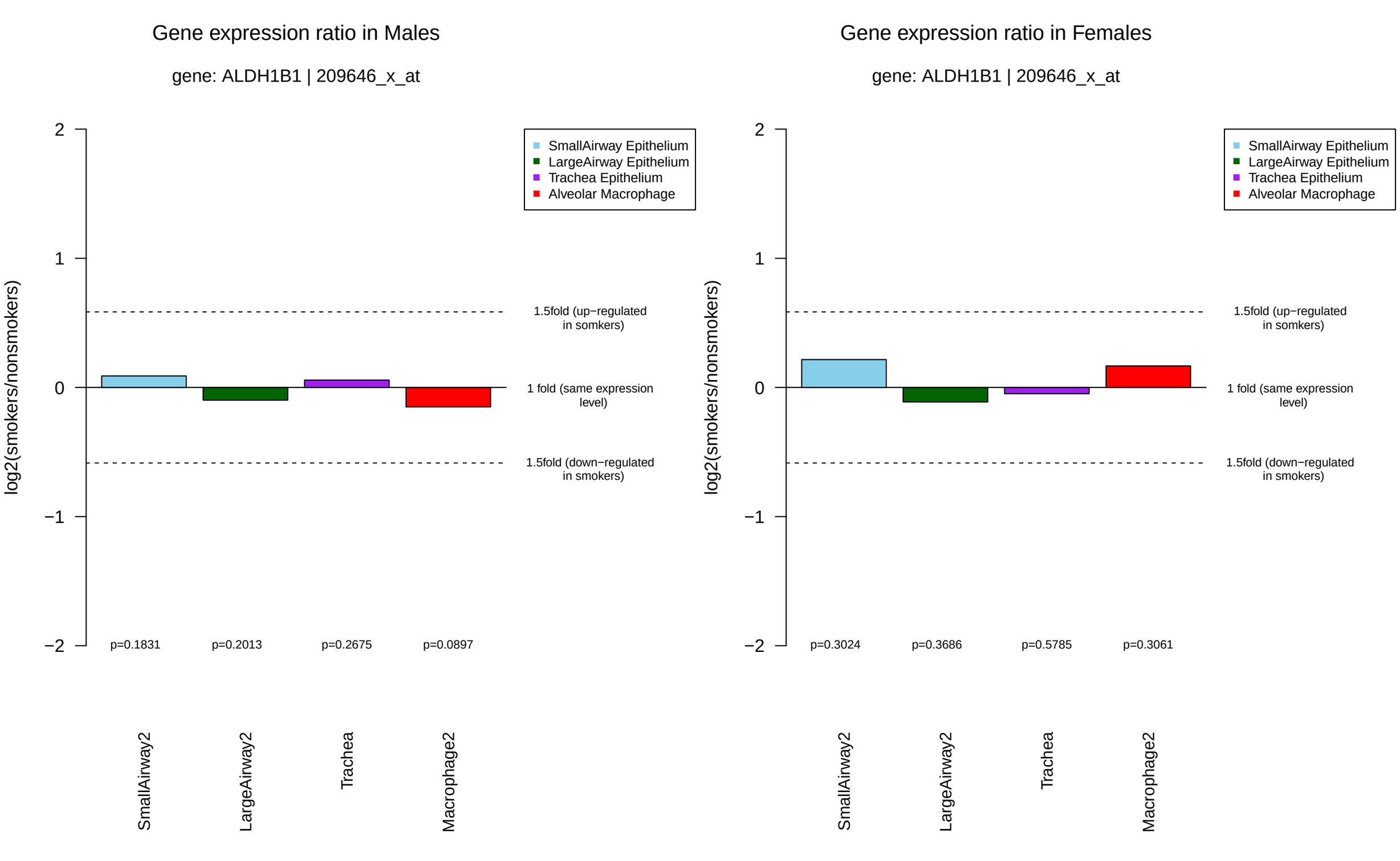
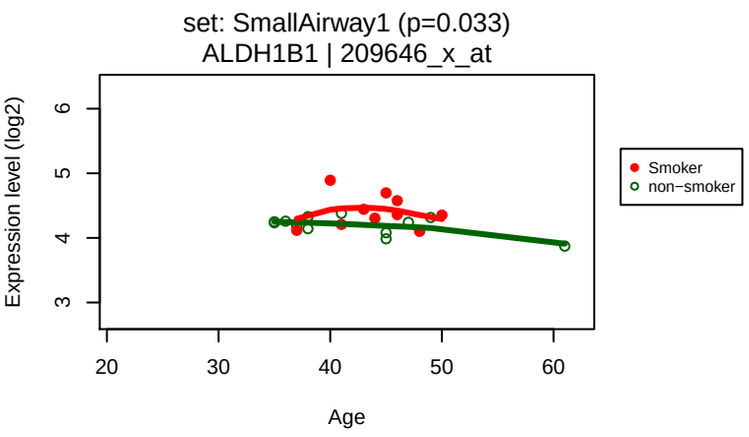
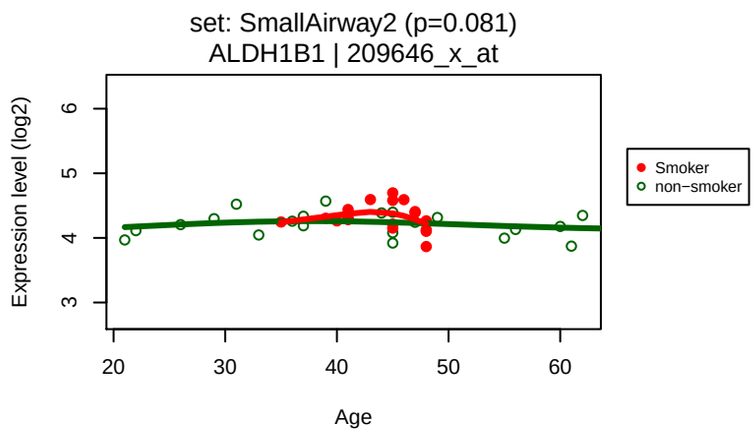
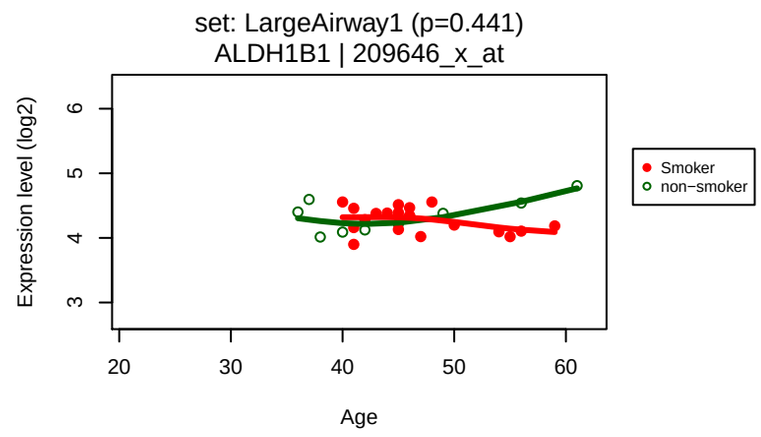

Gene expression of smokers and nonsmokers
gene: ALDH1B1 | 209646_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.033
p=0.081
p=0.9154
p=0.441
p=0.124
p=0.3639
p=0.7031
p=0.5532
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

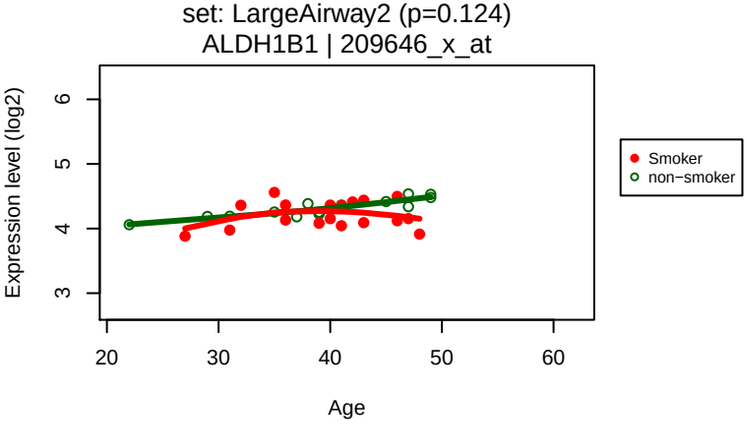

Gene expression ratio (smokers/nonsmokers)
gene: ALDH1B1 | 209646_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.033
p=0.081
p=0.9154
p=0.441
p=0.124
p=0.3639
p=0.7031
p=0.5532
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

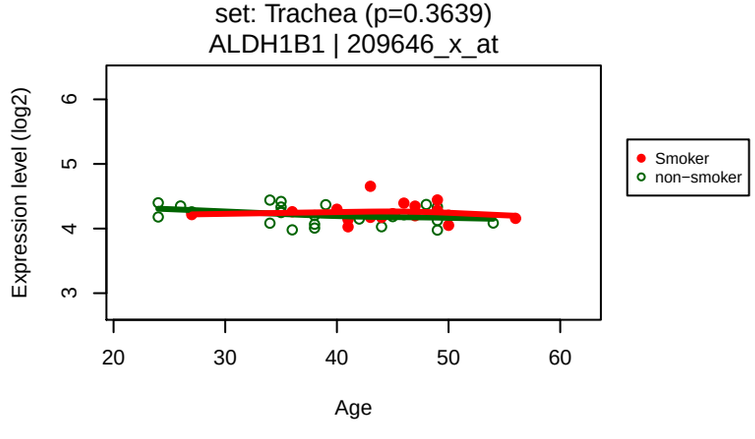

Gene expression ratio in Males
gene: ALDH1B1 | 209646_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1831
p=0.2013
p=0.2675
p=0.0897
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH1B1 | 209646_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3024
p=0.3686
p=0.5785
p=0.3061
SmallAirway2
LargeAirway2
Trachea
Macrophage2

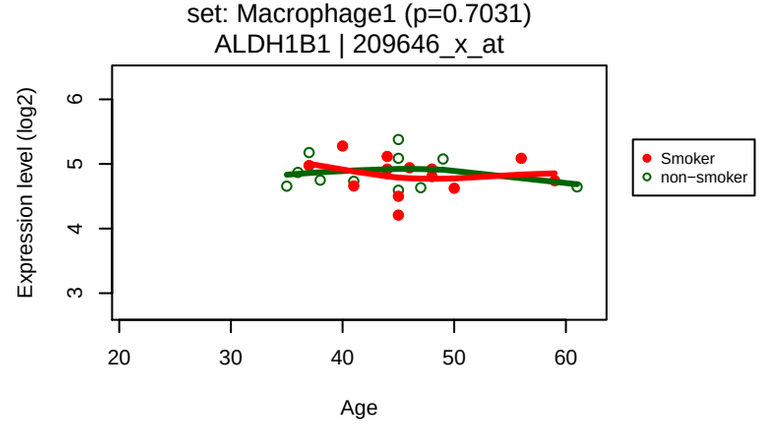

set: SmallAirway1 (p=0.033)
ALDH1B1 | 209646_x_at
Expression level (log2)
Age
Smoker
non−smoker

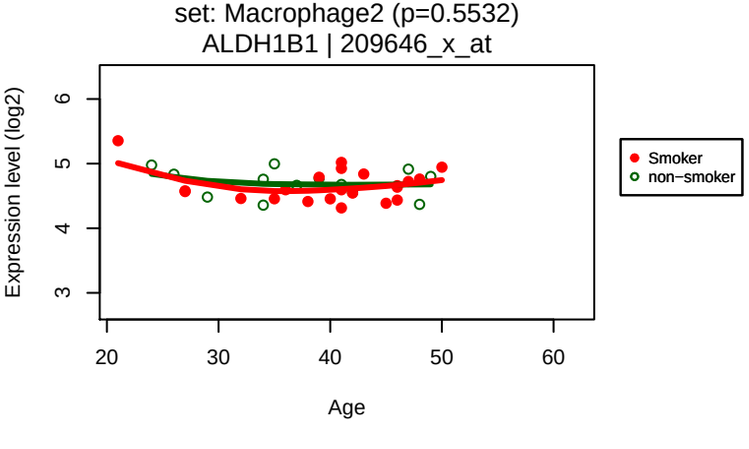

set: SmallAirway2 (p=0.081)
ALDH1B1 | 209646_x_at
Expression level (log2)
Age
Smoker
non−smoker

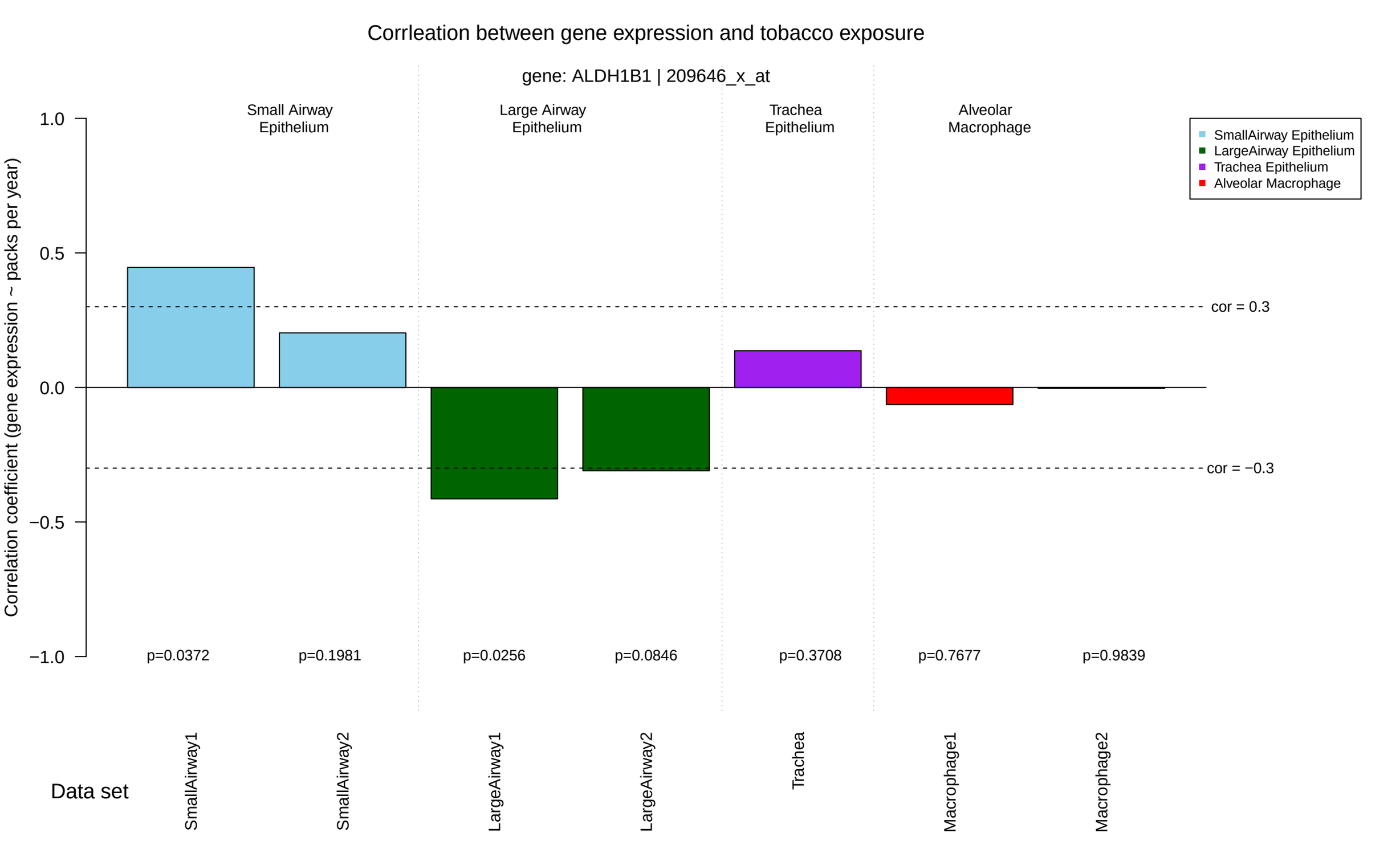

set: LargeAirway1 (p=0.441)
ALDH1B1 | 209646_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.124)
ALDH1B1 | 209646_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3639)
ALDH1B1 | 209646_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7031)
ALDH1B1 | 209646_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5532)
ALDH1B1 | 209646_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH1B1 | 209646_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0372
p=0.1981
p=0.0256
p=0.0846
p=0.3708
p=0.7677
p=0.9839
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2